Ryan Rumana

Emeryville, California | ryan.rumana@gmail.com | Linkedin | Github | Portfolio

WORK EXPERIENCE

Dais — **Tulsa, Oklahoma (Remote)**
Senior AI/ML Engineer — *Nov 2025 - Present*
- Lead ML research and development through architecting projects, driving QA, and mentoring peers.
- Engineered Rust-based multi-tenant context platform utilizing LLM-backed schema induction and temporal RAG.
- Architected an agent-orchestration platform using Cedar engine and Postgres for durable, validated agents.

Axial3D — **Belfast, United Kingdom (Remote)**
Software Engineer — *Jun 2024 - Nov 2025*
- Support and enhance a Vue + Java web app that ingests DICOM scans into our ML 3D-model generation pipeline.
- Contribute towards web app rewrite, adding features, and tightening auth and AWS CloudFormation deployment.

Entegrion — **Durham, North Carolina (Remote)**
Software Engineer — *Jun 2022 - Mar 2024*
- Developed Python/C++ data-processing and visualization tools for viscoelastic coagulation monitor workflows
- Designed and tuned SQL schemas to store and retrieve experiment data efficiently for downstream analysis.

Amergint Technologies — **Colorado Springs, Colorado**
Software Engineer Intern — *May 2022 - Aug 2022*
- Maintained server-side code in C++/Python for custom OS environments, resolving critical bugs.
- Provided hands-on troubleshooting on Linux servers, ensuring swift client support and system reliability.

PROJECTS

Kubernetes Cluster — **Emeryville, California**
Architect & Administrator — *Jul 2024 - Present*
- Run a 6-node Kubernetes cluster with ArgoCD GitOps hosting 70+ services (Nextcloud, Immich, LiteLLM, vLLM).
- Built and manage resilient storage cluster using Rook+Ceph, Postgres, Opensearch, and Valkey.
- Designed an OPNsense network with VLANs, WireGuard, HAProxy edge, and centralized monitoring.

Reverse Game-of-Life — **Emeryville, California**
Solo Research — *Jan 2025 - May 2025*
- Proved reversing Conway's Game of Life on finite boards is NP-complete, then built a working reverse solver.
- Optimized the forward simulator for ~10,000× speedup focusing on performance and correctness.

Stanford EE292D ML on Embedded Systems — **Palo Alto, California**
Research Project — *Apr 2024 - Jul 2024*
- Developed an on-device PyTorch-based person reidentification application for the Raspberry Pi.
- Overcame limited compute power with aggressive quantization and optimized data pipelines
- Invited to present our on-device person reidentification research at Stanford's ACT4AERO workshop.

EDUCATION

Colorado School of Mines — **Golden, Colorado**
MS in Computer Science - Machine Learning — *Graduation Date: May 2023*
BS in Computer Science - Computer Engineering — *Graduation Date: May 2022*

SKILLS

Technologies: Rust, Python, AWS, C++, PyTorch,TypeScript, Postgresql, AWS, Docker, Kubernetes, CI/CD
Expertise: Machine Learing, System Design, Platform Optimization, Software Development, Problem Solving
Interests: Homelab development, Linux customization, Hiking, Competitive swimming, Snowboarding, Triathlons